On The Sharing of New Knowledge through Academic Journals

Lorenzo Ductor Marcel Fafchamps,
Sanjeev Goyal Fulin Guo

Network Science and Economics Conference

October 2026

The sharing of knowledge

- New ideas central to innovation and economic growth (Romer 1990; Aghion and Howitt 1992, Mokyr 2005).
- Diffusion of knowledge is hard to observe
- We rarely see flow of an idea and channel that carried it.
- Academic research is an exception.

Academic journals

- New knowledge is shared mainly through peer-reviewed journals.
- Journals allocate attention and shape careers (Heckman and Moktan 2020).
- Journals form an ecology: partly horizontal and partly hierarchical:
  - **Scope**: field journals vs general journals.
  - **Standing**: top journals vs. non-top journals.
- This paper: where in the journal hierarchy do ideas circulate, and where does outside knowledge enter?

Contributions to literature

- Insularity of economics (Fourcade et al. JEP 2015; Angrist et al. JEL 2020; Truc et al. 2023). **Our finding: economics is similar to other disciplines.**
- Role of top general journals (Card and DellaVigna 2013; Heckman and Moktan 2020). **Our finding: they help in cross-fertilization**
- Outside own discipline intellectual influence (Pinski and Narin 1976; Palacios-Huerta and Volij 2004). **Our finding: general journals most influential**
- Openness to other disciplines (Pieters and Baumgartner 2002; Yan et al. 2013; Wu et al. 2017). **Our finding: field journals more open in most disciplines**

Research questions: within a discipline

- **Cross-fertilization** (general vs. field journals):
  - Are articles in field *A* cited by a **more diverse set of fields** if they are published in a general journal?
- **Badge of honor** (top vs. non-top journals):
  - Do articles in field *A* receive **more citations from field A** if they are published in a top journal?
  - Where is the badge: in **top general** or in **top field** journals?

Research question: across disciplines

- Where does outside knowledge enter? where does inside knowledge flow?
- **Openness**: proportion of references outside own discipline.
- **Influence**: attention a journal receives from other disciplines.
- Possible patterns:
  1. General journals are more open: they **bring ideas** from other disciplines, which then percolate to field journals.
  2. Field journals are more open: they exchange ideas with other disciplines first.

Data

- Web of Science: all articles published between 1970 and 2022.
- **Six** disciplines: economics, mathematics, psychology, political science, sociology and physics.
- Highly rated journals within discipline, plus 8 multidisciplinary journals (including Science, Nature, PNAS).

Discipline	# fields	# journals	# articles	# citations
Economics	9	103	167,951	2,674,426
Mathematics	17	109	146,550	1,510,906
Psychology	14	104	174,937	4,550,462
Political science	12	50	24,812	280,970
Sociology	23	54	25,034	250,523
Physics	9	98	1,267,039	21,968,102
Multidisciplinary	–	8	227,990	6,176,161

Fields and journal types

- **Fields**: communities of authors who cite each other (Leiden algorithm on the author citation network, clusters of at least 100 authors), same rule in every discipline.
  - Economics: 9 fields (energy and environment, labor, finance, trade and growth, micro theory, macro, innovation and IO, marketing, political economy).
- **Field concentration index** of a set of citations: $\mathrm{FCI} = \sum_c s_c^2$, with s_c the share going to field c.
- **Scope**: a journal is **general** if the FCI of its references is ≤ 0.25; otherwise **field**.
- **Standing**: **top** journals are the top 20% by 2015 Article Influence Score within their group.
- Economics, top general: AER, Econometrica, JEL, JPE, QJE, REStud.

Cross-fertilization: measure

- Outcome: FCI_a of the citations article a receives in the 10 years after publication (lower = more diverse audience).
- Articles with few citations mechanically have a high FCI: ZB is the FCI expected if citations were random across fields (simulations).

$$\mathrm{FCI}_a = \alpha + \rho\, ZB + \beta\, General + \theta\, General \times (ZB - \overline{ZB}) + \delta_t + \gamma_c + \epsilon_a$$

- $\rho < 1$: citations to field journal articles stay within the field.
- $\theta > 0$: articles in general journals are cited by more diverse fields.

Finding I: Cross-fertilization across disciplines

Variables	Econ.	Math.	Psycho.	Pol.	Soc.	Phy.
General × ZB	0.155***	0.089**	0.159***	0.114***	0.092	0.073*
	(0.023)	(0.039)	(0.027)	(0.021)	(0.055)	(0.038)
ZB	0.426***	0.193***	0.436***	0.461***	0.400***	0.421***
	(0.016)	(0.010)	(0.019)	(0.019)	(0.026)	(0.018)
General	-0.070***	-0.033**	-0.087***	-0.067***	-0.038**	-0.030***
	(0.010)	(0.015)	(0.015)	(0.009)	(0.017)	(0.010)
Cluster FEs	Yes	Yes	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes	Yes	Yes
N. Obsv.	75,782	74,798	76,492	13,746	20,767	369,372
R-squared	0.250	0.140	0.216	0.325	0.281	0.117

- General journals play a cross-fertilization role in all disciplines.
- Citations in mathematics more concentrated than in other disciplines.

Research question 2: Badge-of-Honor

- We regress the number of citations each article received in its field on:
  - Top general journal dummy.
  - Non-top general journal dummy.
  - Top field journal dummy.

$$E[n_{ac}] = \exp\left(\alpha + \rho\, TopGen_a + \beta\, GenNonTop_a + \varphi\, TopField_a + \gamma_i + \delta_t + \gamma_c\right)$$

- Poisson with **research team** fixed effects γ_i (same authors).
- We report estimated coefficients as **incidence rate ratios**.

Finding 2: Badge-of-Honor

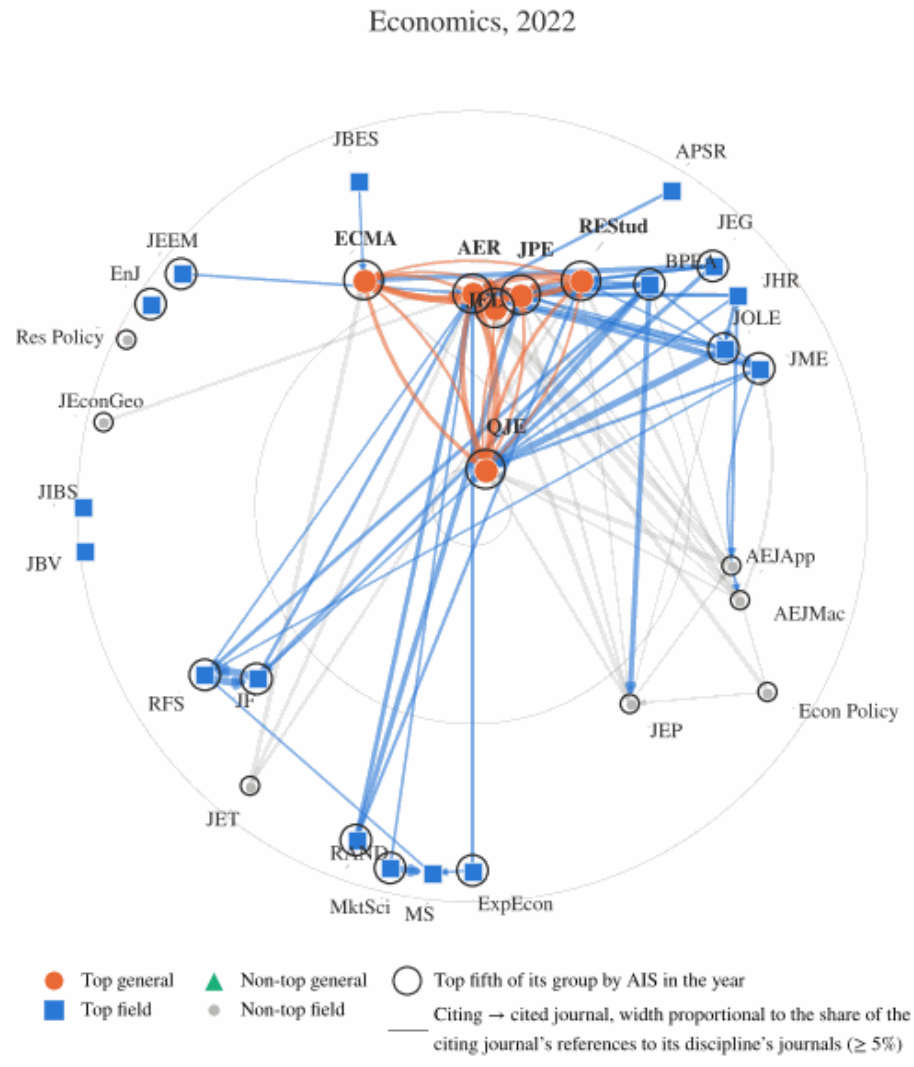
	Econ.	**Math.**	**Psycho.**	**Pol.**	**Soc.**	**Phy.**
	IRR	IRR	IRR	IRR	IRR	IRR
Top general	1.94***	1.35***	1.17	1.68***	1.67***	1.42***
	(0.11)	(0.07)	(0.11)	(0.13)	(0.09)	(0.04)
General non-top	0.86**	1.04	1.28***	0.96	0.91*	1.06
	(0.06)	(0.03)	(0.06)	(0.06)	(0.05)	(0.10)
Top field	1.34***	1.30***	1.42***	0.99	1.35***	1.26***
	(0.09)	(0.06)	(0.10)	(0.06)	(0.08)	(0.04)
p(Top Gen.=Top F.)	<0.001	0.563	0.067	<0.001	0.002	0.003
Author team FE	YES	YES	YES	YES	YES	YES
Observations	30,276	32,349	14,605	5,514	6,666	127,650

Note: IRR >1 (<1) means more (less) citations than non-top field journals.

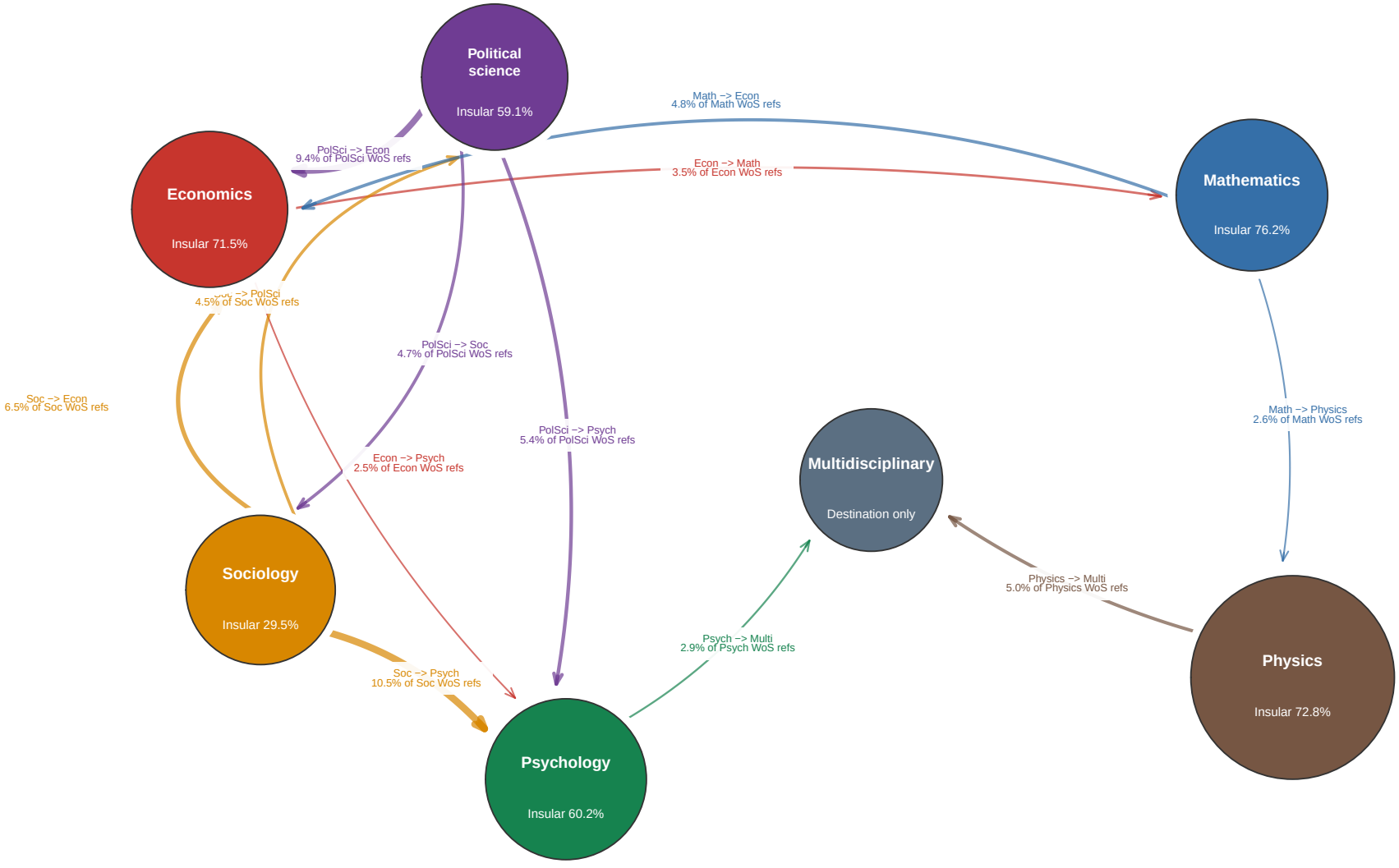
Estimator is Poisson FE.

- Top general journals receive more citations than top field journals in four out of six disciplines.

Citation network of economics journals, 1970-2022



Openness and influence



Finding 3: openness of journals

Openness is the percentage of the references of a journal that go to Web-of-Science journals outside its discipline.

	Econ.	**Math.**	**Psycho.**	**Pol.**	**Soc.**	**Phy.**
Top general	13.9	40.6	40.5	38.2	50.4	16.4
Non-top general	17.6	18.1	35.8	38.7	75.2	18.0
Top field	35.7	30.5	45.4	44.8	70.7	42.1
Non-top	33.0	19.4	37.2	39.6	69.8	20.6
All journals	28.5	23.8	39.8	40.9	70.5	27.2
p (Top G. = Top F.)	$<$0.001	$<$0.001	0.007	$<$0.001	$<$0.001	$<$0.001

- Field journals more open in 5 out of 6 disciplines
- Economics is not especially insular, comparable to physics.
- Economics top general journals are **least open**.

Finding 4: Influence of journals outside their discipline

- Cross-disciplinary Article Influence Score: citations received from other disciplines, weighted by the influence of the citing journals, per article (average article = 1).

	Econ.	**Math.**	**Psycho.**	**Pol.**	**Soc.**	**Phy.**
Top general journals	3.19	3.33	2.19	1.95	4.32	2.60
Non-top general journals	0.57	2.31	0.62	1.04	0.49	1.61
Top field journals	2.05	1.35	1.62	1.05	1.73	0.86
Non-top field journals	0.59	0.68	0.79	0.73	0.57	0.79

- Top general journals largest influence outside their discipline.
- Openness and influence are different.

Concerning Multidisciplinary journals

- Science, Nature, PNAS and five other multidisciplinary journals.
- Cross-fertilization: multidisciplinary journals diffuse ideas more than the general journals of any discipline.
- Badge-of-honor: citations from the discipline of the article (IRR of Science and Nature):
  - High Political science 2.69, physics 2.53, psychology 1.62.
  - Low in economics (0.77), mathematics (0.99) and sociology (1.18).

Summary

Study over 2 millions articles across 6 disciplines over 1970-2022.

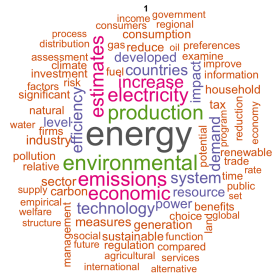
1. General journals perform a **cross-fertilization** role within disciplines.
2. Top journals serve as **badge-of-honor**.
3. Field journals are entry point for ideas from **other disciplines** in 5 out of 6 disciplines.
4. Economics is not insular, but its top general journals are the least open.

Future research:

- Text-mining measures of novelty and disciplinary consensus.

Additional Tables and Figures

Clusters in economics: word clouds



(a) Cluster 1



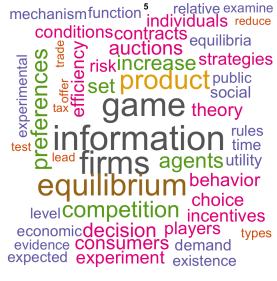
(b) Cluster 2



(c) Cluster 3



(d) Cluster 4



(e) Cluster 5



(f) Cluster 6



(g) Cluster 7



(h) Cluster 8



(i) Cluster 9

Share of JEL codes articles in each cluster

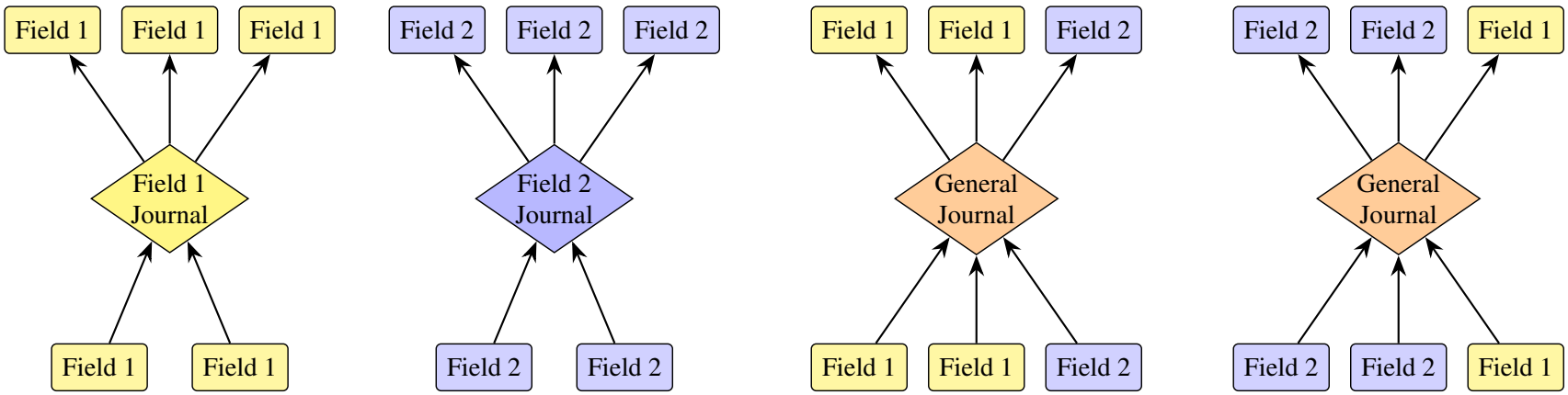
Clusters:	1	2	3	4	5	6	7	8	9
A	0	0	0	0	0	0	0.01	0.01	0.01
B	0	0	0	0	0	0	0.01	0	0.18
C	0.03	0.07	0.05	0.02	0.14	0.16	0.02	0.05	0.01
D	0.09	0.10	0.07	0.17	0.40	0.07	0.07	0.31	0.25
E	0.01	0.03	0.05	0.05	0.02	0.28	0.01	0.01	0
F	0.02	0.02	0.03	0.17	0.01	0.12	0.07	0.01	0.02
G	0.02	0.03	0.58	0.03	0.05	0.09	0.05	0.02	0
H	0.03	0.05	0.01	0.08	0.05	0.03	0.01	0	0.03
I	0.02	0.15	0	0.02	0.02	0.01	0.03	0.02	0.01
J	0.02	0.29	0.01	0.08	0.05	0.04	0.05	0.02	0.07
K	0	0.01	0.01	0.01	0.02	0	0.01	0	0
L	0.12	0.04	0.07	0.08	0.12	0.03	0.21	0.10	0
M	0.01	0.01	0.05	0.01	0.03	0	0.05	0.40	0
N	0	0.01	0	0.01	0	0.01	0.01	0	0
O	0.07	0.09	0.02	0.14	0.03	0.06	0.23	0.02	0
P	0.01	0.01	0	0.02	0	0.01	0.01	0	0.09
Q	0.48	0.05	0.02	0.04	0.03	0.05	0.04	0.01	0.02
R	0.06	0.04	0.01	0.06	0.02	0.01	0.09	0.01	0
Y	0	0	0	0	0	0	0	0	0.27
Z	0	0.01	0	0.01	0.01	0	0.01	0.02	0.05

General journals for at least 10 years - economics

General journal	# years	Top
JOURNAL OF POLITICAL ECONOMY	50	✓
AMERICAN ECONOMIC REVIEW	49	✓
QUARTERLY JOURNAL OF ECONOMICS	47	✓
ECONOMIC INQUIRY	46	
INTERNATIONAL ECONOMIC REVIEW	45	
ECONOMICS LETTERS	44	
EUROPEAN ECONOMIC REVIEW	44	
JOURNAL OF LAW & ECONOMICS	43	
ECONOMIC JOURNAL	42	
OXFORD ECONOMIC PAPERS	42	
REVIEW OF ECONOMICS AND STATISTICS	42	
CANADIAN JOURNAL OF ECONOMICS	38	
ECONOMICA	38	
REVIEW OF ECONOMIC STUDIES	35	✓
SCANDINAVIAN JOURNAL OF ECONOMICS	30	
JOURNAL OF ECONOMIC LITERATURE	29	✓
JOURNAL OF PUBLIC ECONOMICS	27	
JOURNAL OF URBAN ECONOMICS	26	
CAMBRIDGE JOURNAL OF ECONOMICS	25	
JOURNAL OF COMPARATIVE ECONOMICS	24	
OXFORD REVIEW OF ECONOMIC POLICY	23	
JOURNAL OF DEVELOPMENT ECONOMICS	21	
REGIONAL SCIENCE AND URBAN ECONOMICS	21	
ECONOMETRICA	20	✓
JOURNAL OF ECONOMIC SURVEYS	19	
WORLD DEVELOPMENT	19	
JOURNAL OF THE EUROPEAN ECONOMIC ASSOCIATION	15	
JOURNAL OF INDUSTRIAL ECONOMICS	14	
BELL JOURNAL OF ECONOMICS	10	

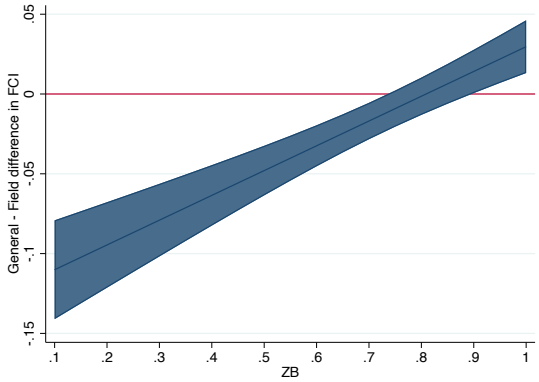
Number of years from 1970 to 2022 in which the journal has a FCI ≤ 0.25. Top: top 20% by 2015 Article Influence Score among general journals.

Cross-fertilization role of general journals: visually

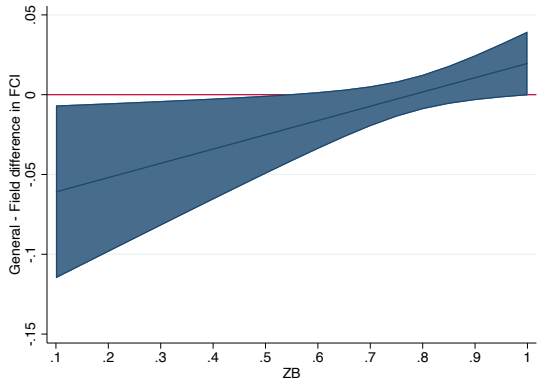


- Each box is an article, labelled by the field of its authors. An arrow indicates a citation.
- Under cross-fertilization, the **composition** of the citing articles changes.

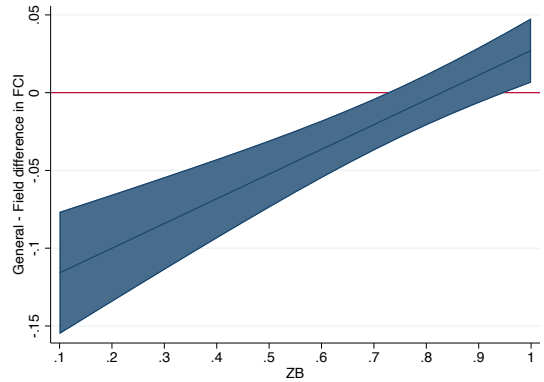
Difference in predicted FCI: general minus field journals



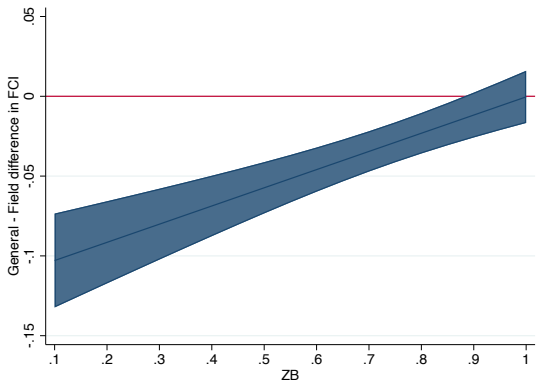
Economics



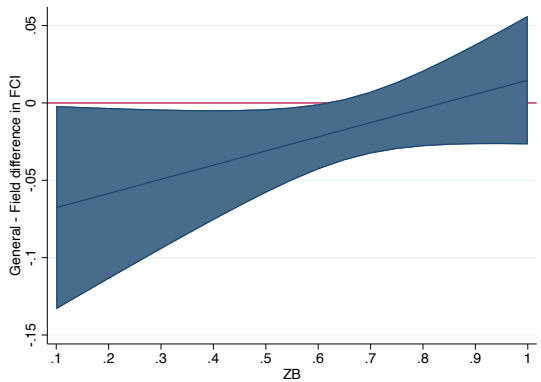
Mathematics



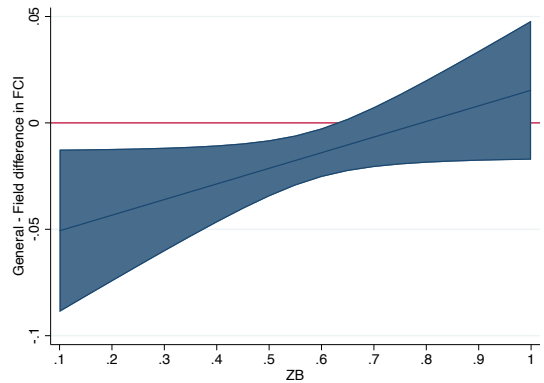
Psychology



Political science



Sociology



Physics

- Below zero: articles in general journals are cited by a more diverse set of fields (95% confidence intervals).

Cross-fertilization: Interpretation

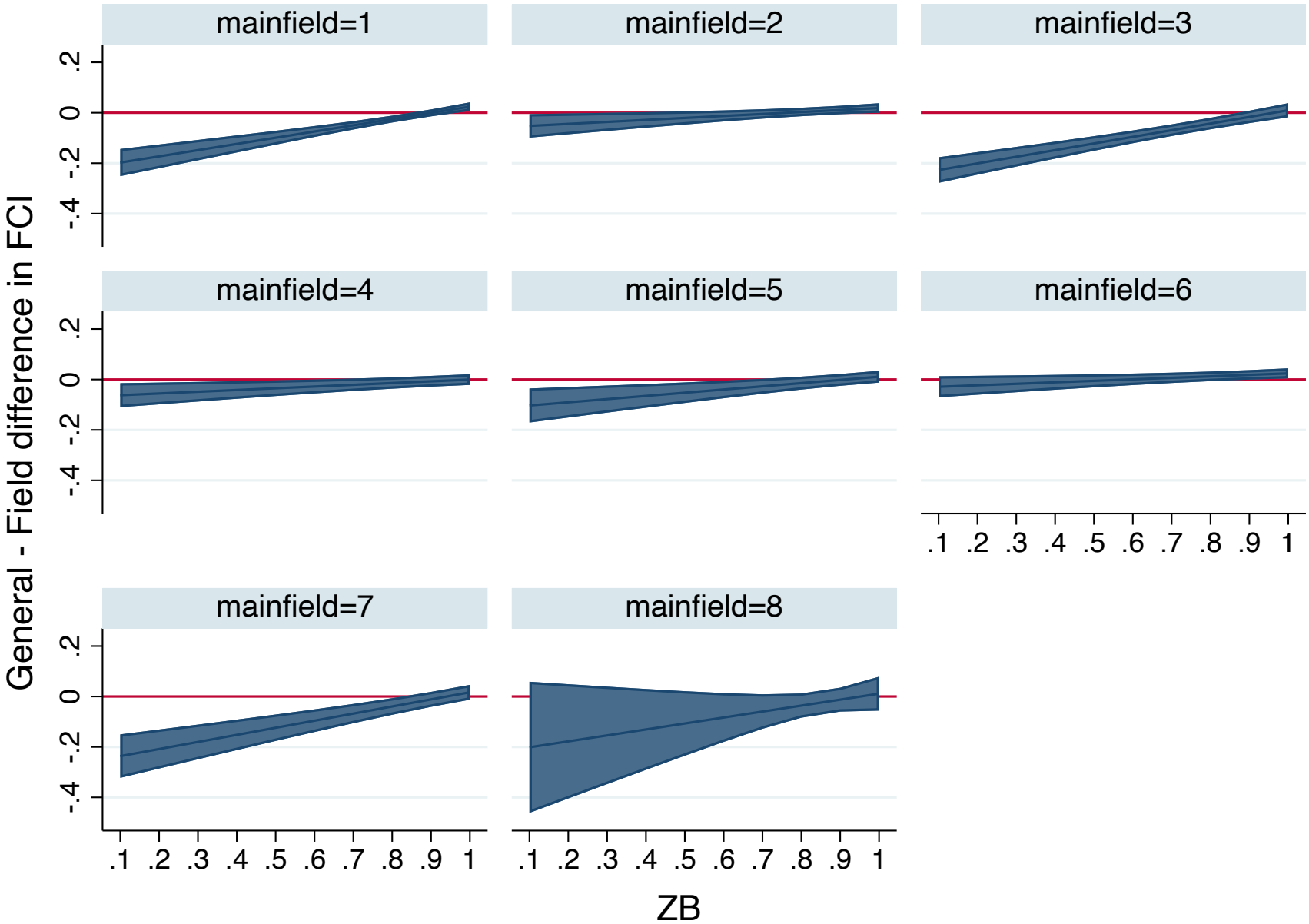
- In all disciplines, articles published in general journals are cited by a more diverse set of fields.
- The gap between general and field journals is larger for well cited articles (low ZB).
- The result holds with research team fixed effects.
- In economics, the gap is present in all fields except labor and macroeconomics.

Cross-fertilization: with research team FE

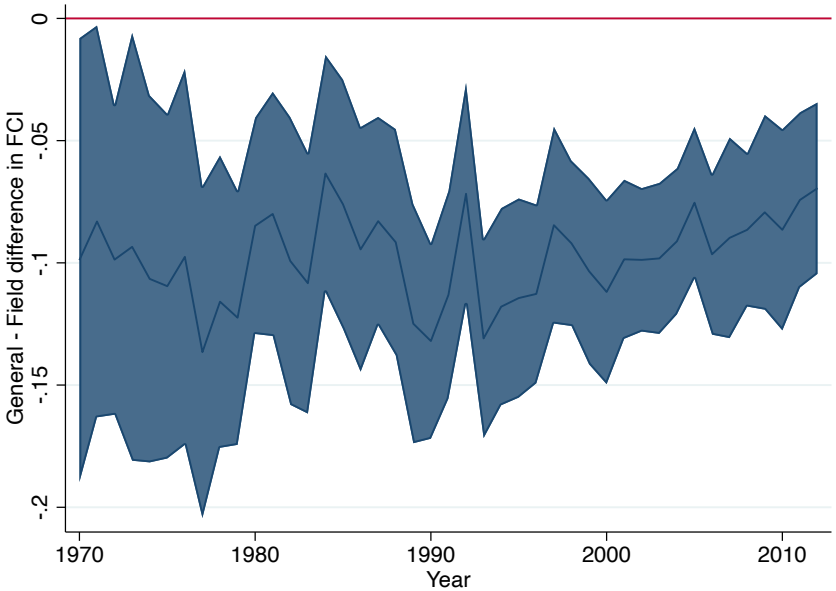
Variables	Econ.	Math.	Psycho.	Pol.	Soc.	Phy.
General × ZB	0.105***	0.051***	0.183***	0.097***	0.096**	0.089***
	(0.023)	(0.017)	(0.037)	(0.022)	(0.041)	(0.033)
ZB	0.467***	0.196***	0.449***	0.452***	0.446***	0.451***
	(0.017)	(0.012)	(0.023)	(0.021)	(0.025)	(0.024)
General journal	-0.061***	-0.020	-0.104***	-0.061***	-0.065***	-0.042***
	(0.013)	(0.013)	(0.023)	(0.012)	(0.022)	(0.012)
Research team FE	Yes	Yes	Yes	Yes	Yes	Yes
Cluster dummies	Yes	Yes	Yes	Yes	Yes	Yes
Year dummies	Yes	Yes	Yes	Yes	Yes	Yes
N. Observations	30,278	32,344	14,605	5,542	5,757	127,644
R-squared	0.641	0.586	0.671	0.644	0.665	0.659

- Inward bias essentially unchanged with research team FE.
- The general-field gap is smaller but negative and significant in all disciplines except mathematics.

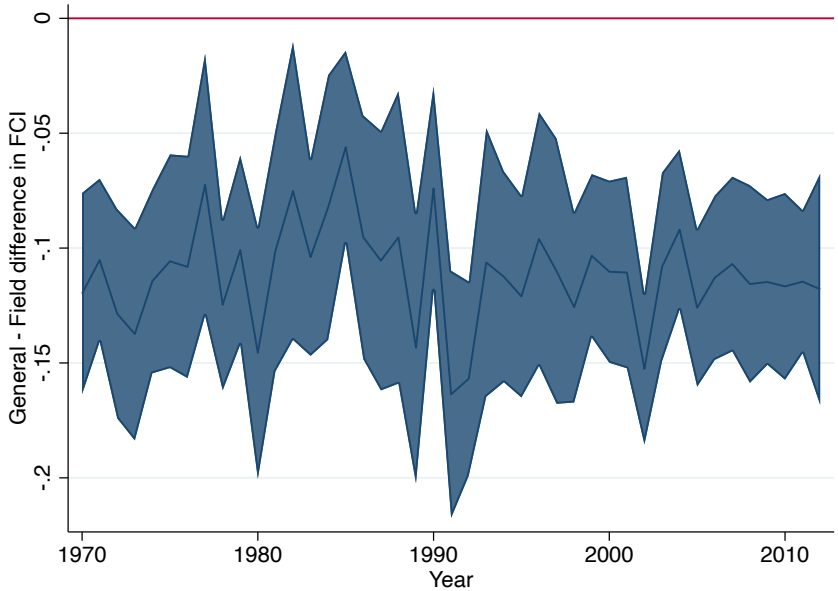
Difference in predicted FCI by field [economics]



Difference in predicted FCI across time [economics and psychology]

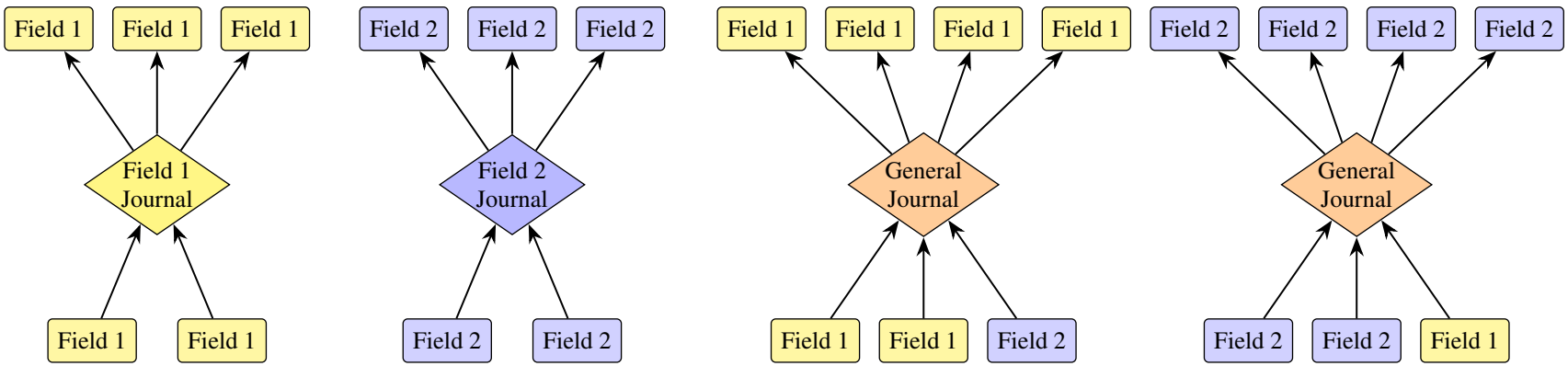


(a) Economics



(d) Psychology

Badge-of-honor role of general journals: visually



- Under the badge of honor, the **number** of citations from the same field changes.
- The composition of the citing articles does not change.

Badge-of-honor: other classifications of top journals

	Econ.	Math.	Psycho.	Pol.	Soc.	Phy.
Prestige lists from the literature						
Top general journals	2.30***	1.79***	1.87***	1.68***	1.75***	1.58***
General non-top journals	0.96	1.06	1.20***	0.99	0.91**	1.09
Top field journals	1.51***	1.15***	1.24***	1.07	1.17**	1.13**
p (top general = top field)	<0.001	<0.001	0.004	<0.001	<0.001	<0.001
Survey ranking without citation data (Van Buskirk et al. 2026)						
Top general journals	1.83***	1.81***	1.34***	–	1.58***	1.49***
General non-top journals	0.78***	1.03	1.09	–	0.83***	1.02
Top field journals	1.23***	1.22***	1.39***	–	1.04	1.15***
p (top general = top field)	<0.001	<0.001	0.620	–	<0.001	<0.001

Note: IRR. Poisson FE with author team, cluster and year FE. Citations from the own field.

Badge-of-honor by decade

		1970s	1980s	1990s	2000s
Economics	Top general	1.47***	1.66***	1.94***	2.22***
	Top field	0.96	1.05	1.42***	1.42***
Mathematics	Top general	1.26	1.26***	1.54***	1.13
	Top field	1.03	1.21***	1.27***	1.44***
Psychology	Top general	1.58***	1.24**	0.93	0.99
	Top field	1.73***	1.47***	1.16	1.48***
Political science	Top general	1.04	1.07	1.28***	2.02***
	Top field	0.90	0.63**	0.80**	0.98
Sociology	Top general	1.26**	1.59***	1.63***	1.71***
	Top field	0.88	1.46**	1.46***	1.39***
Physics	Top general	1.16***	1.23***	1.49***	1.55***
	Top field	1.34***	1.28***	1.31***	1.21***

Note: IRR. Poisson FE by decade of publication; journal categories held fixed.

Citations from the own discipline: multidisciplinary journals

	Econ.	**Math.**	**Psycho.**	**Pol.**	**Soc.**	**Phy.**	**All**
Top general	2.09***	1.54***	1.28***	1.98***	1.93***	1.39***	1.50***
General non-top	0.88***	0.95	1.21***	1.07	0.95	1.03	0.95***
Top field	1.31***	1.29***	1.37***	0.94	1.39***	1.38***	1.31***
Science and Nature	0.77	0.99	1.62***	2.69***	1.18	2.53***	2.38***
Other multidisc.	0.77*	0.69***	1.13	1.73***	0.84	1.31***	1.19**
Observations	33,789	32,842	22,643	5,547	3,414	139,315	249,881

Note: IRR. Poisson FE with author team, cluster and year fixed effects.

Is economics insular?

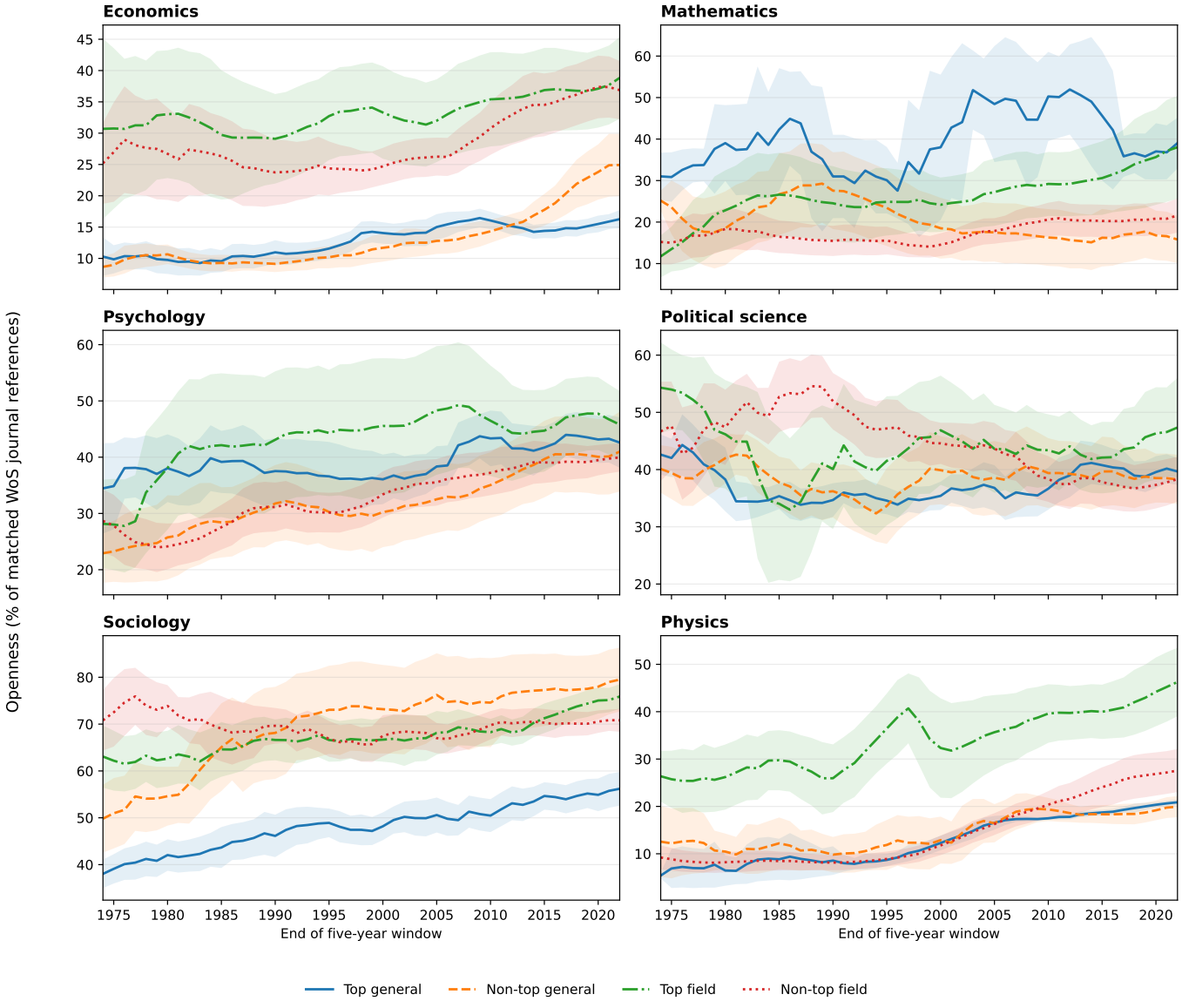
- Proportion of references that go to journals outside the discipline:
  - Economics 28.5%, physics 27.2%, mathematics 23.8%.
  - Psychology 39.8%, political science 40.9%, sociology 70.5%.
- Economics is an important source of knowledge for other disciplines:
  - 9.4% of the references of political science, 6.5% of sociology and 4.8% of mathematics go to economics.
  - Economics sends 3.5% of its references to mathematics, 2.5% to psychology, 1.6% to political science and 1.0% to sociology.
- $\Rightarrow$ Economics is internally concentrated but externally influential.

Openness by field [economics]

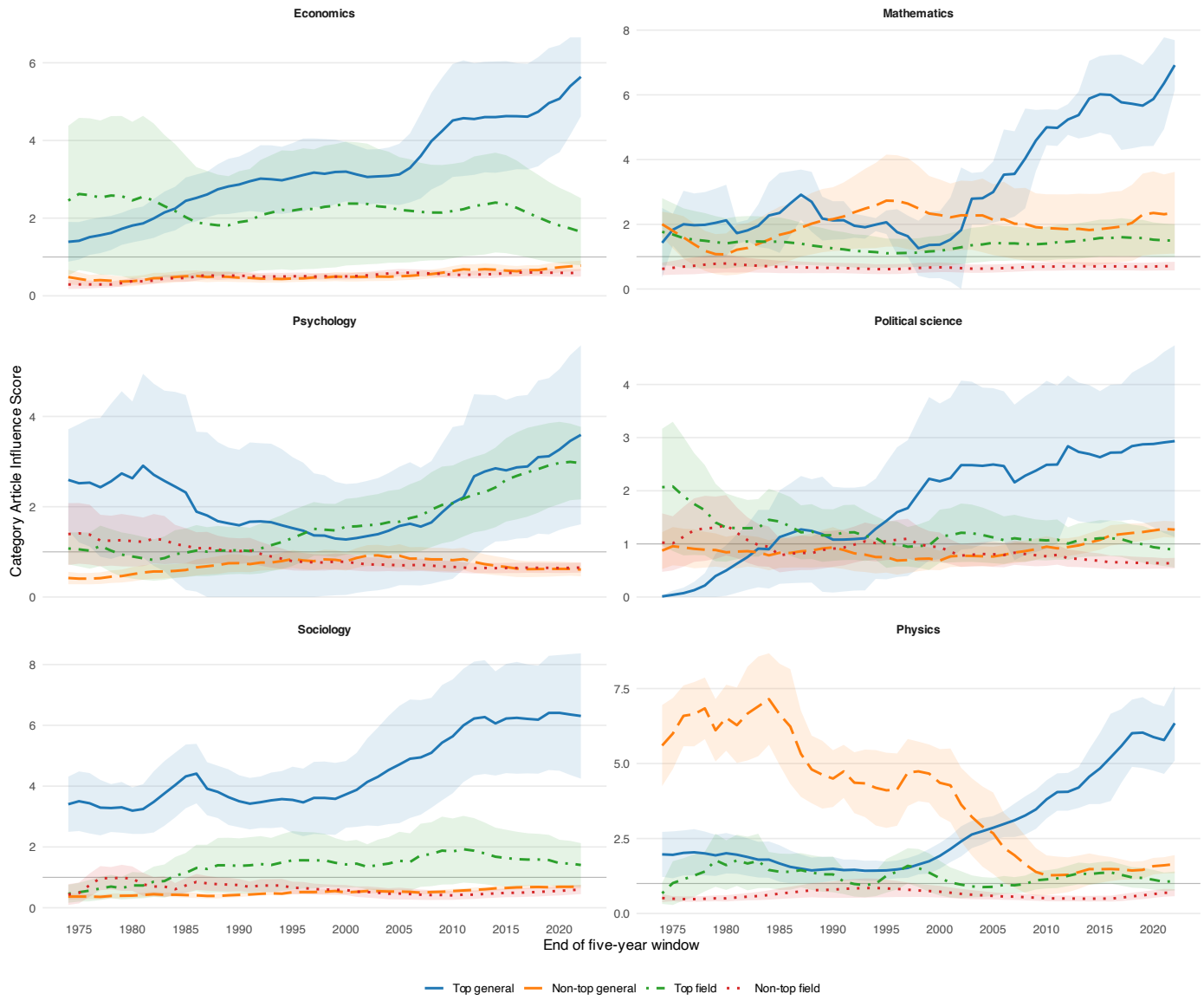
	All journals	Top general journals
Energy and environment	46.0%	14.0%
Labor and education	23.4%	17.1%
Finance	16.8%	10.1%
Trade and growth	15.2%	12.0%
Micro theory	18.3%	15.5%
Macro and time series	14.3%	7.4%
Innovation and IO	50.5%	12.3%
Marketing	78.0%	26.7%
Political economy	49.0%	n.a.

- Large differences in openness across fields.
- In every field, articles in top general journals are less open than the field average.

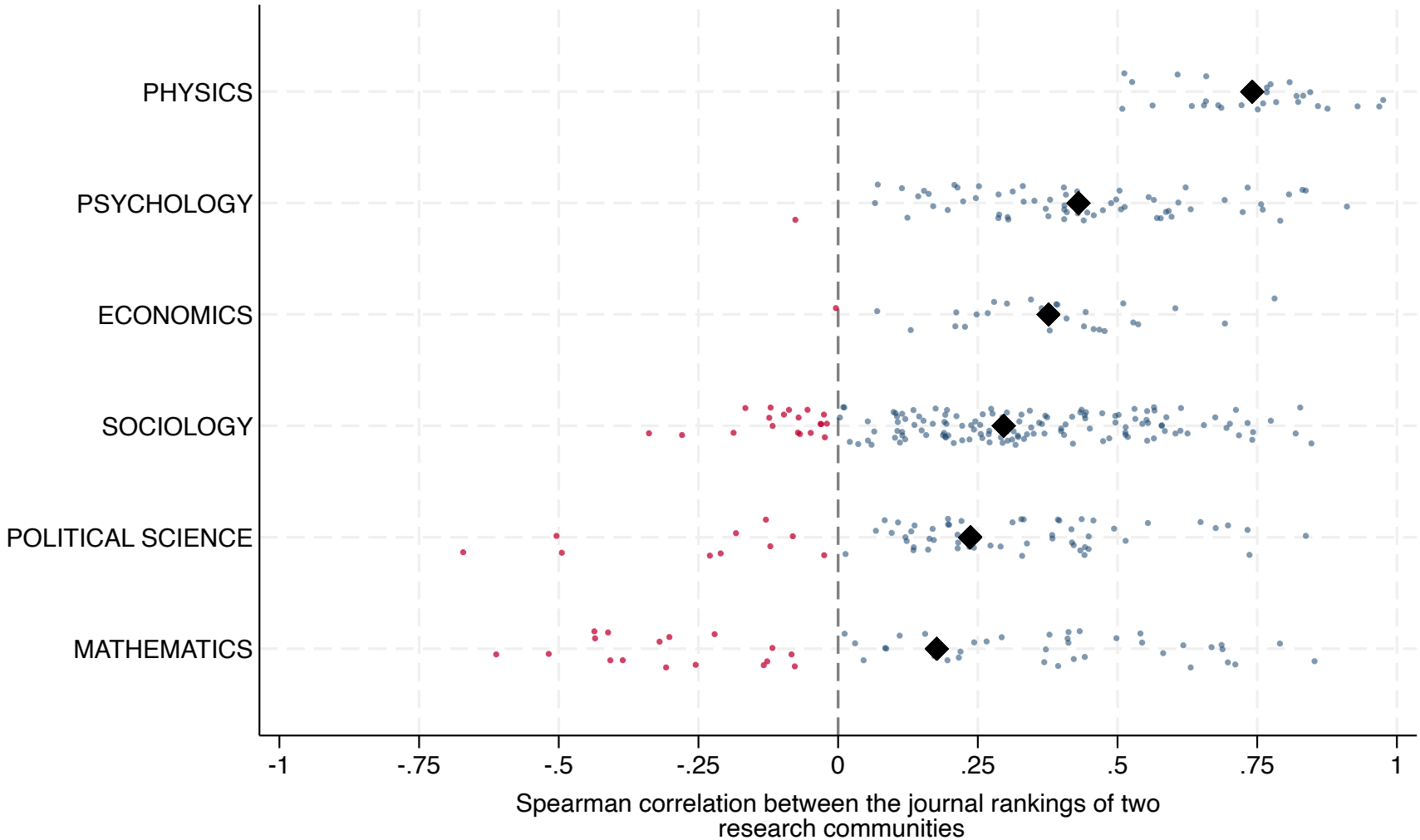
Openness across time



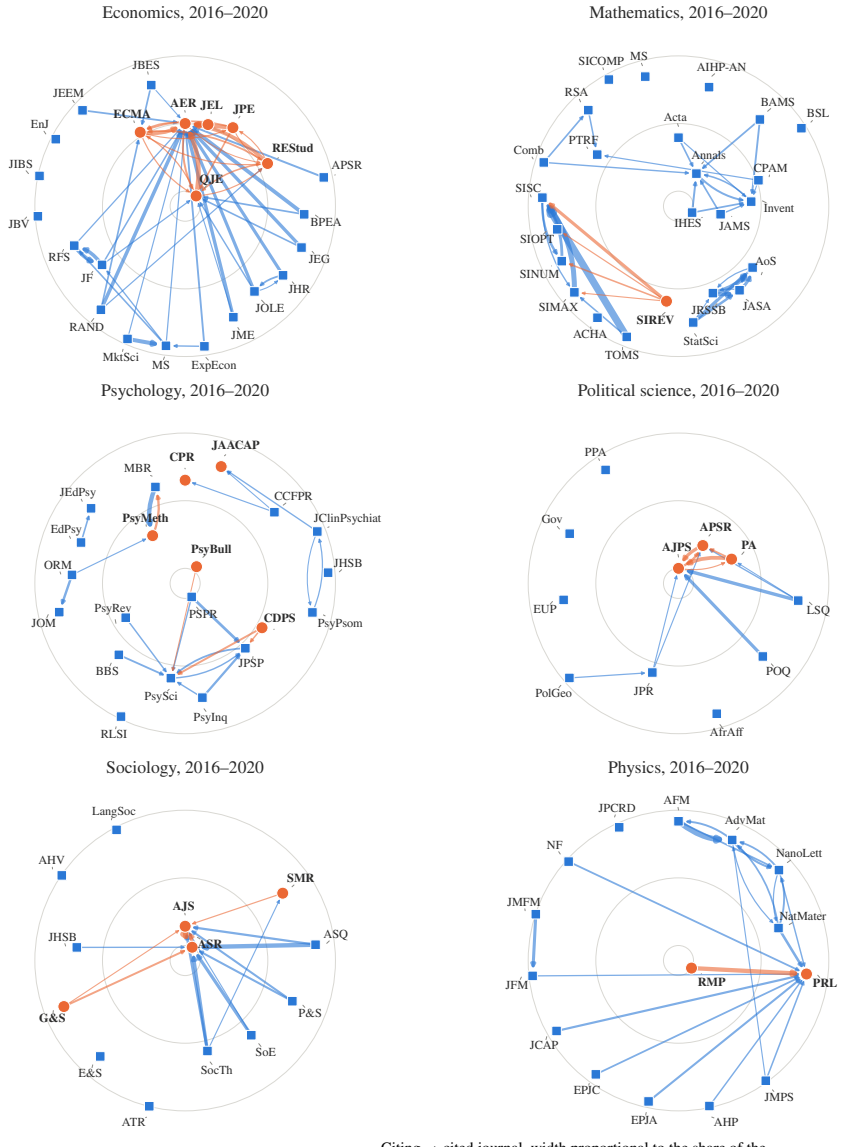
Influence across time



Consensus: pairwise correlations of journal rankings



Citation network of top journals, 2016-2020: all disciplines



Interdisciplinarity of cited field articles

- We examine whether **field** articles with a more **interdisciplinary** bibliography are cited **more** by **general** journals:
  - For each article, we compute the interdisciplinarity ratio of each of the field articles **it cites**.
  - We regress this ratio on a dummy for whether the citing journal is general.
- General journals cite **less** interdisciplinary field articles: negative in all disciplines, significant in economics (-0.050) and physics (-0.032).
- $\Rightarrow$ In economics and physics, new ideas from other disciplines are integrated in field journals before they circulate through general journals.

Openness and consensus

- Consensus: do the fields of a discipline agree on the ranking of journals?
- $\bar{\kappa}$: average Spearman correlation between the journal rankings of two fields.

Discipline	Consensus $\bar{\kappa}$	Openness of top general journals (%)
Physics	0.741	16.4
Psychology	0.430	40.5
Economics	0.376	13.9
Sociology	0.296	50.4
Political science	0.237	38.2
Mathematics	0.176	40.6

- More consensus, less open top general journals: Spearman correlation of -0.49 (six disciplines, not significant).
- Mathematics has two paradigms (pure mathematics and statistics) whose journal rankings are uncorrelated (-0.08).

Changes over time

- **Cross-fertilization**: the general-field gap is stable over five decades (about -0.10 in economics and psychology).
- **Badge-of-honor** increases:
  - Top general journals in economics: IRR from 1.47 (1970s) to 2.22 (2000s).
  - Political science: from 1.04 to 2.02.
- **Influence** of top general journals in economics: from about 1.4 to 5.6 (top field journals: from 2.5 to 1.7).
- **Openness** in economics: top field journals between 16 and 22 points more open than top general journals in every period.
- $\Rightarrow$ To the extent careers follow these returns, economics directs more effort to its least open journals than it once did.